



User: FINAL_STATA 19_OUTPUT_ROBUST

```

1 . set more off

2 .
3 . import excel "D:\SKRIPSI\FINAL_DATASET.xlsx", firstrow clear
   (10 vars, 325 obs)

4 .
5 . * Ubah kode perusahaan menjadi ID numerik
6 . encode Firm_Code, gen(FirmID)

7 .
8 . * Set panel
9 . xtset FirmID Year

   Panel variable: FirmID (unbalanced)
   Time variable: Year, 2011 to 2022, but with gaps
   Delta: 1 unit

10 .
   end of do-file

11 . do "C:\Users\Lenovo\Downloads\02_Winsor.do"

12 . capture which winsor2

13 . if _rc ssc install winsor2

14 .
15 . winsor2 Leverage ESG_Score IM_TONE Asset_Growth Sales_Growth, replace cuts(1 99)

16 .
   end of do-file

17 . do "C:\Users\Lenovo\Downloads\03_Descriptive.do"

18 . summarize Leverage ESG_Score IM_TONE Asset_Growth Sales_Growth

```

Variable	Obs	Mean	Std. dev.	Min	Max
Leverage	325	.7233	1.097269	-1.5992	6.3985
ESG_Score	325	47.9296	19.9098	10.37	87.11
IM_TONE	325	-.0003678	.3093494	-1.203199	.6780605
Asset_Growth	325	.0781846	.1814315	-.26	.91
Sales_Growth	325	.116179	.2380879	-.4229961	1.044436

```

19 .
   end of do-file

20 . do "C:\Users\Lenovo\Downloads\04_Correlation.do"

21 . pwcorr Leverage ESG_Score IM_TONE Asset_Growth Sales_Growth, sig star(0.05)

```

	Leverage	ESG_Score	IM_TONE	Asset_Growth	Sales_Growth
Leverage	1.0000				
ESG_Score	-0.1247*	1.0000			
	0.0245				
IM_TONE	0.1612*	-0.1098*	1.0000		
	0.0036	0.0480			
Asset_Growth	0.0791	-0.0235	0.0962	1.0000	
	0.1546	0.6726	0.0835		

Sales_Growth	-0.0272	0.0228	0.0881	0.4235*	1.0000
	0.6249	0.6822	0.1130	0.0000	

22 .
end of do-file

23 . do "C:\Users\Lenovo\Downloads\05_ModelSelection.do"

24 . *****

25 . * 05_MODEL_SELECTION.do

26 . *****

27 .

28 . * Pastikan panel sudah diset

29 . capture drop FirmID

30 . encode Firm_Code, gen(FirmID)

31 . xtset FirmID Year

Panel variable: FirmID (unbalanced)

Time variable: Year, 2011 to 2022, but with gaps

Delta: 1 unit

32 .

33 . *****

34 . * COMMON EFFECT MODEL (Pooled OLS)

35 . *****

36 .

37 . reg Leverage ESG_Score Asset_Growth Sales_Growth i.Industry_Code i.Year

Source	SS	df	MS	Number of obs	=	325
Model	135.235585	39	3.4675791	F(39, 285)	=	3.88
Residual	254.859855	285	.894245105	Prob > F	=	0.0000
				R-squared	=	0.3467
				Adj R-squared	=	0.2573
Total	390.09544	324	1.20399827	Root MSE	=	.94565

Leverage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ESG_Score	-.0101269	.0039329	-2.57	0.011	-.017868	-.0023857
Asset_Growth	.5003588	.3568423	1.40	0.162	-.202022	1.20274
Sales_Growth	-.4946766	.2938982	-1.68	0.093	-1.073163	.0838099
Industry_Code						
7	-.2814118	.7114465	-0.40	0.693	-1.681768	1.118944
18	.3485786	.2681243	1.30	0.195	-.1791765	.8763337
20	.3879248	.4182421	0.93	0.354	-.4353105	1.21116
23	.109587	.6016255	0.18	0.856	-1.074606	1.29378
31	.057477	.3265775	0.18	0.860	-.5853328	.7002868
40	-.1793926	.7085324	-0.25	0.800	-1.574013	1.215228
42	.7716636	.3970543	1.94	0.053	-.0098673	1.553194
43	.5349977	.3665309	1.46	0.145	-.1864533	1.256449
48	1.70322	.372322	4.57	0.000	.9703706	2.43607
50	-.5190719	.5942998	-0.87	0.383	-1.688846	.6507019
53	.5317535	.3753253	1.42	0.158	-.2070079	1.270515
55	.4718283	.7133081	0.66	0.509	-.9321923	1.875849
56	.274562	.3380322	0.81	0.417	-.3907944	.9399184
58	1.714787	.2789553	6.15	0.000	1.165713	2.263861
64	1.277421	.3501098	3.65	0.000	.5882922	1.96655
71	.1465065	.2774489	0.53	0.598	-.3996024	.6926154
73	.4406707	.3796228	1.16	0.247	-.3065494	1.187891
75	-.1395978	.3417106	-0.41	0.683	-.8121945	.5329988
76	1.330691	.7182883	1.85	0.065	-.0831323	2.744514

78	.1976058	.259316	0.76	0.447	-.3128118	.7080234
85	-.7012441	.7120685	-0.98	0.326	-2.102825	.7003364
90	-.7053812	.9850798	-0.72	0.475	-2.644336	1.233574
92	.2143282	.7102544	0.30	0.763	-1.183682	1.612338
93	-.0359232	.3186887	-0.11	0.910	-.6632054	.591359
96	1.468214	.2852807	5.15	0.000	.9066894	2.029738
Year						
2012	-.2665611	.3396843	-0.78	0.433	-.9351695	.4020472
2013	-.2462299	.347713	-0.71	0.479	-.9306412	.4381813
2014	.1529242	.321581	0.48	0.635	-.4800509	.7858993
2015	-.178566	.3251349	-0.55	0.583	-.8185364	.4614044
2016	-.2028286	.3198773	-0.63	0.527	-.8324503	.4267931
2017	.2916211	.3119203	0.93	0.351	-.3223388	.905581
2018	.1247865	.3120858	0.40	0.690	-.489499	.7390721
2019	.1321771	.307792	0.43	0.668	-.4736569	.7380111
2020	.211416	.3162265	0.67	0.504	-.4110198	.8338518
2021	.3758214	.3010991	1.25	0.213	-.2168387	.9684814
2022	.3274633	.310325	1.06	0.292	-.2833565	.938283
_cons	.6048221	.3286242	1.84	0.067	-.0420164	1.251661

38 . est store POLS

```

39 .
40 . *****
41 . * CHOW TEST (CEM vs FEM)
42 . *****
43 .
44 . reg Leverage ESG_Score Asset_Growth Sales_Growth i.FirmID i.Year

```

Source	SS	df	MS	Number of obs	=	325
Model	222.904808	63	3.53817156	F(63, 261)	=	5.52
Residual	167.190632	261	.640577133	Prob > F	=	0.0000
				R-squared	=	0.5714
				Adj R-squared	=	0.4680
Total	390.09544	324	1.20399827	Root MSE	=	.80036

Leverage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033255	.0040998	0.81	0.418	-.0047475	.0113984
Asset_Growth	.5304697	.3384065	1.57	0.118	-.1358847	1.196824
Sales_Growth	-.4633077	.2765747	-1.68	0.095	-1.00791	.0812941
FirmID						
ABBA	1.468247	.6372372	2.30	0.022	.2134664	2.723027
ABMM	.7297228	.6440299	1.13	0.258	-.538433	1.997879
ACES	-.2972686	.5339107	-0.56	0.578	-1.348589	.7540521
ACST	-.3391122	.6338114	-0.54	0.593	-1.587147	.9089224
ADES	-.357827	.6342893	-0.56	0.573	-1.606803	.8911487
ADHI	1.433945	.8485447	1.69	0.092	-.2369201	3.104809
ADRO	-.0379335	.4164562	-0.09	0.927	-.8579752	.7821083
AMRT	-.1530151	.6265047	-0.24	0.807	-1.386662	1.080632
ANTM	.0737561	.3681975	0.20	0.841	-.6512596	.7987718
ASII	.2037238	.3362089	0.61	0.545	-.4583035	.865751
ASRI	.6244754	.6492329	0.96	0.337	-.6539257	1.902877
BRPT	.4871009	.4379569	1.11	0.267	-.3752777	1.34948
BSDE	.0101935	.3739062	0.03	0.978	-.7260633	.7464503
BUMI	1.095395	.3970698	2.76	0.006	.3135268	1.877263
BYAN	-.4623079	.6538951	-0.71	0.480	-1.749889	.8252736
CPIN	.0398692	.3347845	0.12	0.905	-.6193532	.6990916
EXCL	.9440856	.3466422	2.72	0.007	.2615142	1.626657
GGRM	.0688155	.3618125	0.19	0.849	-.6436276	.7812587
GJTL	.4029684	.628082	0.64	0.522	-.8337846	1.639721

GOOD	.3142853	.6264801	0.50	0.616	-.9193135	1.547884
GOTO	.1315788	.8934208	0.15	0.883	-1.627651	1.890809
HMSF	-.4363281	.4230395	-1.03	0.303	-1.269333	.3966768
ICBP	-.0279386	.351972	-0.08	0.937	-.7210047	.6651276
INCO	-.3346927	.371471	-0.90	0.368	-1.066154	.3967689
INDF	.2758126	.3503015	0.79	0.432	-.4139644	.9655895
INKP	.6168117	.436791	1.41	0.159	-.2432712	1.476895
ISAT	1.55047	.342241	4.53	0.000	.8765648	2.224375
ITMG	-.4174798	.3887872	-1.07	0.284	-1.183039	.348079
JPFA	.3491884	.6305294	0.55	0.580	-.8923838	1.590761
JSMR	1.395103	.3661871	3.81	0.000	.6740461	2.11616
KLBF	-.314654	.3408799	-0.92	0.357	-.9858788	.3565707
LINK	.0884278	.6304794	0.14	0.889	-1.153046	1.329902
LPKR	.3457958	.3633262	0.95	0.342	-.3696279	1.061219
LPPF	1.369852	.3523775	3.89	0.000	.6759872	2.063716
MDKA	.1287362	.4829772	0.27	0.790	-.8222916	1.079764
MTDL	-.3855511	.856312	-0.45	0.653	-2.071711	1.300608
MTEL	.1431978	.8524529	0.17	0.867	-1.535363	1.821758
PGAS	.3273074	.3912924	0.84	0.404	-.4431845	1.097799
POWR	.2673299	.6323085	0.42	0.673	-.9777454	1.512405
PTBA	-.3323666	.3776101	-0.88	0.380	-1.075917	.4111835
PWON	-.0173516	.3906569	-0.04	0.965	-.7865919	.7518888
SMRA	.4827944	.3753987	1.29	0.200	-.2564012	1.22199
TBIG	3.92667	.3774635	10.40	0.000	3.183408	4.669931
TKIM	.5709208	.4727978	1.21	0.228	-.3600629	1.501905
TLKM	.0232731	.3522674	0.07	0.947	-.6703747	.7169209
TOWR	2.177072	.4794731	4.54	0.000	1.232944	3.1212
TPIA	-.1337219	.6338195	-0.21	0.833	-1.381773	1.114329
UNTR	-.2509396	.3519851	-0.71	0.477	-.9440316	.4421524
UNVR	-.0113328	.3798061	-0.03	0.976	-.759207	.7365413
Year						
2012	-.2092016	.2902325	-0.72	0.472	-.7806969	.3622936
2013	-.3179155	.301108	-1.06	0.292	-.9108257	.2749947
2014	.0352383	.2755938	0.13	0.898	-.5074321	.5779086
2015	-.203165	.2798083	-0.73	0.468	-.7541339	.347804
2016	-.2949974	.2753539	-1.07	0.285	-.8371954	.2472006
2017	.1496406	.2684295	0.56	0.578	-.3789225	.6782038
2018	.0347201	.270437	0.13	0.898	-.4977961	.5672362
2019	-.0676025	.2698971	-0.25	0.802	-.5990554	.4638504
2020	.0335849	.280091	0.12	0.905	-.5179408	.5851105
2021	.0869237	.2673853	0.33	0.745	-.4395833	.6134306
2022	.1318389	.2800487	0.47	0.638	-.4196034	.6832813
_cons	.2425007	.3295248	0.74	0.462	-.406365	.8913663

45 .

46 . testparm i.FirmID

```

( 1) 2.FirmID = 0
( 2) 3.FirmID = 0
( 3) 4.FirmID = 0
( 4) 5.FirmID = 0
( 5) 6.FirmID = 0
( 6) 7.FirmID = 0
( 7) 8.FirmID = 0
( 8) 9.FirmID = 0
( 9) 10.FirmID = 0
(10) 11.FirmID = 0
(11) 12.FirmID = 0
(12) 13.FirmID = 0
(13) 14.FirmID = 0
(14) 15.FirmID = 0
(15) 16.FirmID = 0
(16) 17.FirmID = 0

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(17) 18.FirmID = 0
(18) 19.FirmID = 0
(19) 20.FirmID = 0
(20) 21.FirmID = 0
(21) 22.FirmID = 0
(22) 23.FirmID = 0
(23) 24.FirmID = 0
(24) 25.FirmID = 0
(25) 26.FirmID = 0
(26) 27.FirmID = 0
(27) 28.FirmID = 0
(28) 29.FirmID = 0
(29) 30.FirmID = 0
(30) 31.FirmID = 0
(31) 32.FirmID = 0
(32) 33.FirmID = 0
(33) 34.FirmID = 0
(34) 35.FirmID = 0
(35) 36.FirmID = 0
(36) 37.FirmID = 0
(37) 38.FirmID = 0
(38) 39.FirmID = 0
(39) 40.FirmID = 0
(40) 41.FirmID = 0
(41) 42.FirmID = 0
(42) 43.FirmID = 0
(43) 44.FirmID = 0
(44) 45.FirmID = 0
(45) 46.FirmID = 0
(46) 47.FirmID = 0
(47) 48.FirmID = 0
(48) 49.FirmID = 0
(49) 50.FirmID = 0
```

F(49, 261) =	6.43
Prob > F =	0.0000

```

47 .
48 . *****
49 . * FIXED EFFECT MODEL
50 . *****
51 .
52 . xtreg Leverage ESG_Score Asset_Growth Sales_Growth i.Year, fe

```

Fixed-effects (within) regression	Number of obs	=	325
Group variable: FirmID	Number of groups	=	50

R-squared:	Obs per group:
Within = 0.0557	min = 1
Between = 0.0180	avg = 6.5
Overall = 0.0182	max = 12

corr(u_i, Xb) = -0.0492	F(14, 261)	=	1.10
	Prob > F	=	0.3579

Leverage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033255	.0040998	0.81	0.418	-.0047475	.0113984
Asset_Growth	.5304697	.3384065	1.57	0.118	-.1358847	1.196824
Sales_Growth	-.4633077	.2765747	-1.68	0.095	-1.00791	.0812941
Year						
2012	-.2092016	.2902325	-0.72	0.472	-.7806969	.3622936
2013	-.3179155	.301108	-1.06	0.292	-.9108257	.2749947
2014	.0352383	.2755938	0.13	0.898	-.5074321	.5779086
2015	-.203165	.2798083	-0.73	0.468	-.7541339	.347804
2016	-.2949974	.2753539	-1.07	0.285	-.8371954	.2472006
2017	.1496406	.2684295	0.56	0.578	-.3789225	.6782038
2018	.0347201	.270437	0.13	0.898	-.4977961	.5672362
2019	-.0676025	.2698971	-0.25	0.802	-.5990554	.4638504
2020	.0335849	.280091	0.12	0.905	-.5179408	.5851105
2021	.0869237	.2673853	0.33	0.745	-.4395833	.6134306
2022	.1318389	.2800487	0.47	0.638	-.4196034	.6832813
_cons	.5988931	.2629035	2.28	0.024	.0812113	1.116575
sigma_u	.79335937					
sigma_e	.80036063					
rho	.49560706	(fraction of variance due to u_i)				

F test that all u_i=0: F(49, 261) = 6.43

Prob > F = 0.0000

53 . est store FE

54 .

55 . *****

56 . * RANDOM EFFECT MODEL

57 . *****

58 .

59 . xtreg Leverage ESG_Score Asset_Growth Sales_Growth i.Industry_Code i.Year, re

Random-effects GLS regression

Group variable: FirmID

Number of obs = 325

Number of groups = 50

R-squared:

Within = 0.0296

Between = 0.5284

Overall = 0.3467

Obs per group:

min = 1

avg = 6.5

max = 12

corr(u_i, X) = 0 (assumed)

Wald chi2(39) = 151.23

Prob > chi2 = 0.0000

Leverage	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ESG_Score	-.0101269	.0039329	-2.57	0.010	-.0178351	-.0024186
Asset_Growth	.5003588	.3568423	1.40	0.161	-.1990393	1.199757
Sales_Growth	-.4946766	.2938982	-1.68	0.092	-1.070707	.0813533
Industry_Code						
7	-.2814118	.7114465	-0.40	0.692	-1.675821	1.112998
18	.3485786	.2681243	1.30	0.194	-.1769353	.8740925
20	.3879248	.4182421	0.93	0.354	-.4318146	1.207664
23	.109587	.6016255	0.18	0.855	-1.069577	1.288751
31	.057477	.3265775	0.18	0.860	-.5826031	.697557
40	-.1793926	.7085324	-0.25	0.800	-1.568091	1.209305
42	.7716636	.3970543	1.94	0.052	-.0065485	1.549876
43	.5349977	.3665309	1.46	0.144	-.1833896	1.253385
48	1.70322	.372322	4.57	0.000	.9734827	2.432958
50	-.5190719	.5942998	-0.87	0.382	-1.683878	.6457343

53	.5317535	.3753253	1.42	0.157	-.2038707	1.267378
55	.4718283	.7133081	0.66	0.508	-.92623	1.869887
56	.274562	.3380322	0.81	0.417	-.3879689	.9370929
58	1.714787	.2789553	6.15	0.000	1.168044	2.261529
64	1.277421	.3501098	3.65	0.000	.5912187	1.963624
71	.1465065	.2774489	0.53	0.597	-.3972833	.6902963
73	.4406707	.3796228	1.16	0.246	-.3033763	1.184718
75	-.1395978	.3417106	-0.41	0.683	-.8093383	.5301426
76	1.330691	.7182883	1.85	0.064	-.0771284	2.73851
78	.1976058	.259316	0.76	0.446	-.3106443	.7058559
85	-.7012441	.7120685	-0.98	0.325	-2.096873	.6943845
90	-.7053812	.9850798	-0.72	0.474	-2.636102	1.22534
92	.2143282	.7102544	0.30	0.763	-1.177745	1.606401
93	-.0359232	.3186887	-0.11	0.910	-.6605416	.5886952
96	1.468214	.2852807	5.15	0.000	.909074	2.027354
Year						
2012	-.2665611	.3396843	-0.78	0.433	-.9323302	.3992079
2013	-.2462299	.347713	-0.71	0.479	-.9277348	.4352749
2014	.1529242	.321581	0.48	0.634	-.4773629	.7832114
2015	-.178566	.3251349	-0.55	0.583	-.8158187	.4586867
2016	-.2028286	.3198773	-0.63	0.526	-.8297765	.4241193
2017	.2916211	.3119203	0.93	0.350	-.3197315	.9029737
2018	.1247865	.3120858	0.40	0.689	-.4868904	.7364635
2019	.1321771	.307792	0.43	0.668	-.4710842	.7354384
2020	.211416	.3162265	0.67	0.504	-.4083766	.8312086
2021	.3758214	.3010991	1.25	0.212	-.2143219	.9659647
2022	.3274633	.310325	1.06	0.291	-.2807626	.9356892
_cons	.6048221	.3286242	1.84	0.066	-.0392696	1.248914
sigma_u	0					
sigma_e	.80036063					
rho	0	(fraction of variance due to u_i)				

60 . est store RE

```

61 .
62 . *****
63 . * HAUSMAN TEST
64 . *****
65 .
66 . hausman FE RE, sigmamore

```

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
ESG_Score	.0033255	-.0101269	.0134523	.002828
Asset_Growth	.5304697	.5003588	.0301109	.1803659
Sales_Growth	-.4633077	-.4946766	.0313689	.1428594
Year				
2012	-.2092016	-.2665611	.0573595	.0469724
2013	-.3179155	-.2462299	-.0716856	.0752687
2014	.0352383	.1529242	-.1176859	.0511325
2015	-.203165	-.178566	-.0245989	.0598646
2016	-.2949974	-.2028286	-.0921688	.059354
2017	.1496406	.2916211	-.1419805	.0573899
2018	.0347201	.1247865	-.0900665	.0685603
2019	-.0676025	.1321771	-.1997796	.0833956
2020	.0335849	.211416	-.1778311	.0975615
2021	.0869237	.3758214	-.2888977	.0956355
2022	.1318389	.3274633	-.1956243	.1148162

b = Consistent under H0 and Ha; obtained from xtreg.

B = Inconsistent under H_a , efficient under H_0 ; obtained from xtreg.

Test of H_0 : Difference in coefficients not systematic

$$\chi^2(14) = (b-B)'[(V_b-V_B)^{-1}](b-B) = 87.62$$

Prob > $\chi^2 = 0.0000$

```

67 .
    end of do-file

68 . do "C:\Users\Lenovo\Downloads\06_Classicassumption.do"

69 . *****
70 . * 06_UJI_ASUMSI_KLASIK.do
71 . *****
72 .
73 . * Pastikan panel sudah diset
74 . capture drop FirmID

75 . encode Firm_Code, gen(FirmID)

76 .
77 . xtset FirmID Year

    Panel variable: FirmID (unbalanced)
    Time variable: Year, 2011 to 2022, but with gaps
    Delta: 1 unit

78 .
79 . *****
80 . * 1. UJI MULTIKOLINEARITAS
81 . *****
82 .
83 . capture drop ESG_IM

84 . gen ESG_IM = ESG_Score * IM_TONE

85 .
86 . reg Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth i.Industry_Code i.Year

```

Source	SS	df	MS	Number of obs	=	325
Model	137.571845	41	3.35541085	F(41, 283)	=	3.76
Residual	252.523595	283	.892309523	Prob > F	=	0.0000
				R-squared	=	0.3527
				Adj R-squared	=	0.2589
Total	390.09544	324	1.20399827	Root MSE	=	.94462

Leverage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ESG_Score	-.0106997	.0039524	-2.71	0.007	-.0184796	-.0029198
IM_TONE	.7598154	.5428911	1.40	0.163	-.3088016	1.828432
ESG_IM	-.0102424	.0105505	-0.97	0.332	-.0310098	.010525
Asset_Growth	.4528216	.3586542	1.26	0.208	-.2531468	1.15879
Sales_Growth	-.4890651	.2946886	-1.66	0.098	-1.069125	.0909946
Industry_Code						
7	-.4370211	.7190973	-0.61	0.544	-1.852479	.9784369
18	.3473334	.2726582	1.27	0.204	-.189362	.8840289
20	.3844646	.4199943	0.92	0.361	-.4422445	1.211174
23	.0862908	.6013498	0.14	0.886	-1.097395	1.269977
31	.0889699	.3320343	0.27	0.789	-.5646005	.7425403
40	-.2960047	.7114495	-0.42	0.678	-1.696409	1.1044
42	.819217	.4174106	1.96	0.051	-.0024065	1.640841
43	.5325717	.3674006	1.45	0.148	-.190613	1.255756

48	1.731009	.3726329	4.65	0.000	.9975253	2.464493
50	-.6600047	.6009111	-1.10	0.273	-1.842827	.5228178
53	.5295692	.3794316	1.40	0.164	-.2172971	1.276435
55	.4527905	.7128227	0.64	0.526	-.9503168	1.855898
56	.2058212	.3403453	0.60	0.546	-.4641084	.8757507
58	1.663972	.2813183	5.91	0.000	1.11023	2.217714
64	1.170923	.3558748	3.29	0.001	.4704251	1.87142
71	.1357081	.2777036	0.49	0.625	-.4109187	.6823349
73	.4073444	.3808033	1.07	0.286	-.3422219	1.156911
75	-.2198858	.3454922	-0.64	0.525	-.8999465	.4601749
76	1.257401	.7196315	1.75	0.082	-.1591084	2.673911
78	.1555508	.2614982	0.59	0.552	-.3591775	.6702791
85	-.8846156	.7234193	-1.22	0.222	-2.308581	.5393499
90	-.9662962	.9991979	-0.97	0.334	-2.933099	1.000507
92	.1877157	.7099316	0.26	0.792	-1.209701	1.585132
93	-.0121228	.3194114	-0.04	0.970	-.6408464	.6166008
96	1.399803	.2903329	4.82	0.000	.8283167	1.971289
Year						
2012	-.2255521	.3402665	-0.66	0.508	-.8953265	.4442223
2013	-.2080359	.3482966	-0.60	0.551	-.8936167	.4775449
2014	.1930459	.3239669	0.60	0.552	-.4446448	.8307365
2015	-.139628	.3279119	-0.43	0.671	-.7850837	.5058278
2016	-.194904	.325285	-0.60	0.550	-.8351891	.4453811
2017	.2753433	.3180556	0.87	0.387	-.3507116	.9013982
2018	.099734	.3179489	0.31	0.754	-.5261108	.7255787
2019	.1591558	.3132765	0.51	0.612	-.457492	.7758035
2020	.2789658	.3224686	0.87	0.388	-.3557756	.9137071
2021	.4045029	.3060333	1.32	0.187	-.1978875	1.006893
2022	.366132	.3138249	1.17	0.244	-.2515952	.9838592
_cons	.6294764	.3294645	1.91	0.057	-.0190356	1.277988

87 .

88 . vif

Variable	VIF	1/VIF
ESG_Score	2.25	0.444742
IM_TONE	10.24	0.097644
ESG_IM	10.82	0.092463
Asset_Growth	1.54	0.650419
Sales_Growth	1.79	0.559460
Industry_C~e		
7	1.15	0.868143
18	3.70	0.270593
20	1.35	0.738562
23	1.20	0.830171
31	1.77	0.565692
40	1.13	0.886907
42	1.71	0.585251
43	1.75	0.571997
48	1.51	0.663019
50	1.20	0.831383
53	1.86	0.536298
55	1.13	0.883494
56	1.50	0.666552
58	1.97	0.507254
64	1.51	0.662952
71	1.92	0.520545
73	1.42	0.703182
75	1.55	0.646840
76	1.15	0.866854
78	2.33	0.428637
85	1.17	0.857800

90	1.12	0.896501
92	1.12	0.890704
93	1.64	0.611287
96	2.10	0.476243
Year		
2012	2.09	0.478367
2013	2.31	0.432603
2014	2.41	0.414507
2015	2.58	0.388284
2016	2.74	0.365435
2017	2.81	0.356297
2018	2.99	0.334191
2019	3.17	0.315156
2020	3.27	0.305995
2021	4.14	0.241268
2022	3.87	0.258297
Mean VIF	2.41	

```

89 .
90 . *****
91 . * 2. UJI HETEROSKEDASTISITAS
92 . *****
93 .
94 . xtreg Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth i.Year, fe

```

```

Fixed-effects (within) regression      Number of obs   =       325
Group variable: FirmID                 Number of groups =       50

```

```

R-squared:                               Obs per group:
  Within = 0.0583                        min =          1
  Between = 0.0342                       avg =         6.5
  Overall = 0.0240                       max =         12

```

```

corr(u_i, Xb) = -0.0263                  F(16, 259)      =       1.00
                                           Prob > F        =       0.4553

```

Leverage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033159	.0041375	0.80	0.424	-.0048316	.0114634
IM_TONE	-.244855	.4991257	-0.49	0.624	-1.227716	.7380061
ESG_IM	.0072133	.0098648	0.73	0.465	-.0122121	.0266388
Asset_Growth	.5064739	.3430992	1.48	0.141	-.1691452	1.182093
Sales_Growth	-.4953509	.2800914	-1.77	0.078	-1.046897	.0561955
Year						
2012	-.2091833	.2917349	-0.72	0.474	-.7836576	.365291
2013	-.3277479	.3026514	-1.08	0.280	-.9237187	.2682228
2014	.007505	.2789857	0.03	0.979	-.5418641	.5568742
2015	-.2356858	.2834834	-0.83	0.407	-.7939115	.32254
2016	-.3444238	.2823257	-1.22	0.224	-.9003698	.2115223
2017	.0983867	.2765049	0.36	0.722	-.4460972	.6428707
2018	-.0138523	.2782601	-0.05	0.960	-.5617925	.5340879
2019	-.1145875	.2763147	-0.41	0.679	-.6586969	.4295218
2020	-.0073827	.2863754	-0.03	0.979	-.5713034	.5565379
2021	.0431933	.2731574	0.16	0.874	-.4946988	.5810854
2022	.0952039	.2842264	0.33	0.738	-.4644849	.6548926
_cons	.6461982	.2701571	2.39	0.017	.1142142	1.178182
sigma_u	.78763684					
sigma_e	.80235178					
rho	.49074605	(fraction of variance due to u_i)				

F test that all u_i=0: F(49, 259) = 6.19

Prob > F = 0.0000

95 .
96 . xttest3

Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

H0: $\sigma(i)^2 = \sigma^2$ for all i

chi2 (50) = 344457.70
Prob > chi2 = 0.0000

97 .
98 . *****
99 . * 3. UJI AUTOKORELASI
100 . *****
101 .
102 . xtserial Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth

Wooldridge test for autocorrelation in panel data

H0: no first order autocorrelation
F(1, 32) = 0.692
Prob > F = 0.4117

103 .
end of do-file

104 . do "C:\Users\Lenovo\Downloads\07_FinalRegression.do"

105 . *****
106 . * DO 7 - REGRESI FINAL PANEL DATA
107 . *****
108 .
109 . est clear

110 .
111 . *****
112 . * MODEL 1 (H1)
113 . *****
114 .
115 . xtreg Leverage ESG_Score Asset_Growth Sales_Growth i.Year, fe vce(cluster FirmID)

Fixed-effects (within) regression Number of obs = 325
Group variable: FirmID Number of groups = 50

R-squared: Obs per group:
 Within = 0.0557 min = 1
 Between = 0.0180 avg = 6.5
 Overall = 0.0182 max = 12

 F(14, 49) = 1.73
corr(u_i, Xb) = -0.0492 Prob > F = 0.0799

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033255	.0031521	1.05	0.297	-.003009	.0096599
Asset_Growth	.5304697	.4729397	1.12	0.267	-.4199381	1.480878
Sales_Growth	-.4633077	.2767275	-1.67	0.100	-1.019413	.092797
Year						
2012	-.2092016	.2148469	-0.97	0.335	-.6409527	.2225494
2013	-.3179155	.2722521	-1.17	0.249	-.8650267	.2291957

2014	.0352383	.4043087	0.09	0.931	-.7772504	.847727
2015	-.203165	.3475245	-0.58	0.561	-.9015416	.4952117
2016	-.2949974	.3655973	-0.81	0.424	-1.029693	.439698
2017	.1496406	.1389579	1.08	0.287	-.1296057	.428887
2018	.0347201	.233218	0.15	0.882	-.433949	.5033892
2019	-.0676025	.197492	-0.34	0.734	-.4644774	.3292724
2020	.0335849	.2843665	0.12	0.906	-.5378709	.6050407
2021	.0869237	.2976587	0.29	0.772	-.5112438	.6850912
2022	.1318389	.3444069	0.38	0.704	-.5602727	.8239506
_cons	.5988931	.1249545	4.79	0.000	.3477876	.8499987
sigma_u	.79335937					
sigma_e	.80036063					
rho	.49560706	(fraction of variance due to u_i)				

```

116 .
117 . est store Model1

118 .
119 . *****
120 . * MODEL 2 (H2 - Moderasi)
121 . *****
122 .
123 . capture drop ESG_IM

124 . gen ESG_IM = ESG_Score * IM_TONE

125 .
126 . xtreg Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth i.Year, fe vce(cluster FirmID)

```

```

Fixed-effects (within) regression      Number of obs   =      325
Group variable: FirmID                 Number of groups =       50

```

```

R-squared:                               Obs per group:
  Within = 0.0583                        min =          1
  Between = 0.0342                       avg =         6.5
  Overall = 0.0240                       max =         12

```

```

                                F(16, 49)      =      1.41
corr(u_i, Xb) = -0.0263          Prob > F       =      0.1742

```

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033159	.0031037	1.07	0.291	-.0029213	.0095531
IM_TONE	-.244855	.4007279	-0.61	0.544	-1.050148	.5604379
ESG_IM	.0072133	.0069093	1.04	0.302	-.0066715	.0210981
Asset_Growth	.5064739	.4554706	1.11	0.272	-.4088285	1.421776
Sales_Growth	-.4953509	.2794905	-1.77	0.083	-1.057008	.0663063
Year						
2012	-.2091833	.2098464	-1.00	0.324	-.6308855	.2125189
2013	-.3277479	.2811388	-1.17	0.249	-.8927175	.2372216
2014	.007505	.4101036	0.02	0.985	-.816629	.831639
2015	-.2356858	.366114	-0.64	0.523	-.9714194	.5000479
2016	-.3444238	.3956377	-0.87	0.388	-1.139488	.45064
2017	.0983867	.1680779	0.59	0.561	-.2393784	.4361518
2018	-.0138523	.2662444	-0.05	0.959	-.5488905	.5211859
2019	-.1145875	.2211284	-0.52	0.607	-.5589617	.3297867
2020	-.0073827	.2965081	-0.02	0.980	-.6032381	.5884726
2021	.0431933	.3150646	0.14	0.892	-.5899527	.6763393
2022	.0952039	.3532934	0.27	0.789	-.6147658	.8051736

_cons	.6461982	.1348293	4.79	0.000	.3752486	.9171479
sigma_u	.78763684					
sigma_e	.80235178					
rho	.49074605	(fraction of variance due to u_i)				

```

127 .
128 . est store Model2

129 .
130 . *****
131 . * INSTALL ESTTAB (JIKA BELUM ADA)
132 . *****
133 .
134 . capture which esttab

135 . if _rc ssc install estout

136 .
137 . *****
138 . * CEK MODEL
139 . *****
140 .
141 . est dir

```

Name	Command	Dependent variable	Number of param.	Title
<u>Model1</u>	xtreg	Leverage	16	
<u>Model2</u>	xtreg	Leverage	18	

```

142 .
143 . *****
144 . * EXPORT KE WORD
145 . *****
146 .
147 . esttab Model1 Model2 using "D:\SKRIPSI\Hasil_Regresi.rtf", replace ///
    > b(%9.4f) se(%9.4f) star(* 0.10 ** 0.05 *** 0.01) ///
    > stats(r2_w N, labels("R² Within" "Observasi")) ///
    > title(Hasil Regresi Data Panel)
    (output written to D:\SKRIPSI\Hasil_Regresi.rtf)

148 .
    end of do-file

149 . do "C:\Users\Lenovo\Downloads\08_RobustnessCheck.do"

150 . *-----
151 . * MODEL 1 (HASIL UTAMA)
152 . *-----

```

153 . xtreg Leverage ESG_Score Asset_Growth Sales_Growth i.Year, fe vce(cluster FirmID)

```

Fixed-effects (within) regression      Number of obs   =       325
Group variable: FirmID                 Number of groups =       50

R-squared:                             Obs per group:
    Within = 0.0557                     min =          1
    Between = 0.0180                    avg =         6.5
    Overall = 0.0182                    max =         12

                                F(14, 49)      =       1.73
corr(u_i, Xb) = -0.0492                Prob > F       =       0.0799

```

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033255	.0031521	1.05	0.297	-.003009	.0096599
Asset_Growth	.5304697	.4729397	1.12	0.267	-.4199381	1.480878
Sales_Growth	-.4633077	.2767275	-1.67	0.100	-1.019413	.092797
Year						
2012	-.2092016	.2148469	-0.97	0.335	-.6409527	.2225494
2013	-.3179155	.2722521	-1.17	0.249	-.8650267	.2291957
2014	.0352383	.4043087	0.09	0.931	-.7772504	.847727
2015	-.203165	.3475245	-0.58	0.561	-.9015416	.4952117
2016	-.2949974	.3655973	-0.81	0.424	-1.029693	.439698
2017	.1496406	.1389579	1.08	0.287	-.1296057	.428887
2018	.0347201	.233218	0.15	0.882	-.433949	.5033892
2019	-.0676025	.197492	-0.34	0.734	-.4644774	.3292724
2020	.0335849	.2843665	0.12	0.906	-.5378709	.6050407
2021	.0869237	.2976587	0.29	0.772	-.5112438	.6850912
2022	.1318389	.3444069	0.38	0.704	-.5602727	.8239506
_cons	.5988931	.1249545	4.79	0.000	.3477876	.8499987
sigma_u	.79335937					
sigma_e	.80036063					
rho	.49560706	(fraction of variance due to u_i)				

```

154 .
155 . *-----
156 . * MODEL 2 (HASIL UTAMA)
157 . *-----
158 . capture drop ESG_IM

159 . gen ESG_IM = ESG_Score*IM_TONE

160 .
161 . xtreg Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth i.Year, fe vce(cluster FirmID)

```

```

Fixed-effects (within) regression      Number of obs   =       325
Group variable: FirmID                 Number of groups =       50

R-squared:                             Obs per group:
    Within = 0.0583                     min =          1
    Between = 0.0342                    avg =         6.5
    Overall = 0.0240                    max =         12

                                F(16, 49)      =       1.41
corr(u_i, Xb) = -0.0263                Prob > F       =       0.1742

```

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0033159	.0031037	1.07	0.291	-.0029213	.0095531
IM_TONE	-.244855	.4007279	-0.61	0.544	-1.050148	.5604379
ESG_IM	.0072133	.0069093	1.04	0.302	-.0066715	.0210981
Asset_Growth	.5064739	.4554706	1.11	0.272	-.4088285	1.421776
Sales_Growth	-.4953509	.2794905	-1.77	0.083	-1.057008	.0663063
Year						
2012	-.2091833	.2098464	-1.00	0.324	-.6308855	.2125189
2013	-.3277479	.2811388	-1.17	0.249	-.8927175	.2372216
2014	.007505	.4101036	0.02	0.985	-.816629	.831639
2015	-.2356858	.366114	-0.64	0.523	-.9714194	.5000479
2016	-.3444238	.3956377	-0.87	0.388	-1.139488	.45064
2017	.0983867	.1680779	0.59	0.561	-.2393784	.4361518
2018	-.0138523	.2662444	-0.05	0.959	-.5488905	.5211859
2019	-.1145875	.2211284	-0.52	0.607	-.5589617	.3297867
2020	-.0073827	.2965081	-0.02	0.980	-.6032381	.5884726
2021	.0431933	.3150646	0.14	0.892	-.5899527	.6763393
2022	.0952039	.3532934	0.27	0.789	-.6147658	.8051736
_cons	.6461982	.1348293	4.79	0.000	.3752486	.9171479
sigma_u	.78763684					
sigma_e	.80235178					
rho	.49074605	(fraction of variance due to u_i)				

```

162 .
163 . *-----
164 . * ROBUSTNESS CHECK (TANPA YEAR FIXED EFFECTS)
165 . *-----
166 .
167 . * Model 1 Robustness
168 . xtreg Leverage ESG_Score Asset_Growth Sales_Growth, fe vce(cluster FirmID)

```

```

Fixed-effects (within) regression      Number of obs   =      325
Group variable: FirmID                 Number of groups =       50

R-squared:                             Obs per group:
    Within = 0.0221                      min =          1
    Between = 0.0065                     avg  =         6.5
    Overall = 0.0000                     max  =        12

                                F(3, 49)      =       1.33
corr(u_i, Xb) = -0.1785              Prob > F      =       0.2744

```

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0048528	.0031311	1.55	0.128	-.0014393	.0111449
Asset_Growth	.6325899	.4847113	1.31	0.198	-.341474	1.606654
Sales_Growth	-.341659	.2560049	-1.33	0.188	-.8561202	.1728022
_cons	.4809422	.1683509	2.86	0.006	.1426285	.819256
sigma_u	.80079098					
sigma_e	.79782626					
rho	.50185455	(fraction of variance due to u_i)				

```

169 .
170 . * Model 2 Robustness
171 . xtreg Leverage ESG_Score IM_TONE ESG_IM Asset_Growth Sales_Growth, fe vce(cluster FirmID)

```

```

Fixed-effects (within) regression      Number of obs   =      325
Group variable: FirmID                 Number of groups =       50

```

```

R-squared:                             Obs per group:
    Within = 0.0269                     min =          1
    Between = 0.0247                    avg =         6.5
    Overall = 0.0030                     max =         12

```

```

corr(u_i, Xb) = -0.1067                F(5, 49)         =       2.21
                                         Prob > F         =     0.0687

```

(Std. err. adjusted for 50 clusters in FirmID)

Leverage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ESG_Score	.0044118	.0031006	1.42	0.161	-.0018191	.0106427
IM_TONE	-.2635736	.3660002	-0.72	0.475	-.9990786	.4719313
ESG_IM	.0085553	.0055788	1.53	0.132	-.0026558	.0197663
Asset_Growth	.5908075	.4730112	1.25	0.218	-.359744	1.541359
Sales_Growth	-.3770237	.2514322	-1.50	0.140	-.8822956	.1282482
_cons	.5152728	.1621012	3.18	0.003	.1895181	.8410274
sigma_u	.79064803					
sigma_e	.79881081					
rho	.49486456	(fraction of variance due to u_i)				

```

172 .
    end of do-file

```

```

173 .

```